CustardFin

A Cost-Optimized Hybrid Edge-Cloud Media Streaming and Community Discovery Platform

Cloud Computing Project Report

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Candidate Declaration

We hereby declare that the project entitled **"CustardFin: A Cost-Optimized Hybrid Edge-Cloud Media Streaming and Community Discovery Platform"** submitted by us in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA) is our own original work. All sources of assistance, libraries, frameworks, and literature utilized have been properly cited and acknowledged.

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Abstract

Traditional video streaming forces a hard choice: self-hosted servers (like Jellyfin and Plex) keep videos private at home but cannot be easily shared over the internet due to home router firewalls (CGNAT), while commercial platforms (like YouTube and AWS) store all videos in the cloud, leading to high monthly storage and bandwidth bills.

CustardFin solves this problem with a simple rule: *videos stay on the owner's home computer, while the cloud handles only the catalog and stream coordination.* Marking a video "Public" uploads only a small JSON manifest ($\leq$ 3 KB) containing the title and streaming playlist—never the original video file. This reduces cloud operational costs by **95.89%** compared to traditional cloud video hosting.

The system uses pre-encoded 4-second HTTP Live Streaming (HLS) with GPU hardware acceleration (NVIDIA NVENC, with automatic CPU fallback), keeping host CPU usage below 5%. Home nodes connect to the cloud using an outbound-only reverse tunnel (FRP), completely bypassing residential ISP CGNAT without opening any router ports. To protect home upload speeds, an *Origin Shield* cloud cache pulls each requested video chunk once and serves all other viewers directly from the cloud. A 10-stream cap per node prevents home network overload, while local users on the same Wi-Fi enjoy instant, direct playback with under 4 ms latency. CustardFin is built with Go and Next.js using 100% free and open-source tools, and has been verified across local containers, local cloud emulators, and Google Cloud Platform (GCP).

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CHAPTER 1

Introduction and Problem Statement

1.1 Overview and Motivation

Video streaming represents the majority of global internet traffic today. Currently, video platforms fall into two main categories, each with serious drawbacks:

1. **Centralized Public Cloud Platforms (e.g., YouTube, AWS):** Users upload video files to cloud storage servers, which transcode and stream them globally. While this allows easy sharing and public discovery, it is very expensive. Cloud storage (₹ 1.932 per GB/month on AWS S3) and data transfer fees (₹ 7.56 per GB) lead to high monthly bills. In addition, creators lose control of their raw files to third-party platforms.
2. **Self-Hosted Home Media Servers (e.g., Jellyfin, Plex):** Users run media software on a home PC or home server, streaming videos stored on local hard drives. This gives full privacy and zero cloud hosting costs. However, these servers remain locked to the local home network. They cannot be shared with friends easily because home internet providers block incoming connections, and they lack any public discovery feed.

CustardFin bridges this gap by combining the personal ownership of Jellyfin with the community sharing of YouTube. The core principle of CustardFin is simple:

> *"Video files stay permanently on the owner's home computer;*
> *the cloud handles only the catalog and stream coordination."*

In CustardFin, a user hosts their library on an existing PC (the *Home Edge Node*). Marking a video as **Public** does not upload the large video file to the cloud. Instead, it uploads only a small JSON manifest ($\leq$ 3 KB) containing the title and streaming playlist. Viewers discover videos through a modern web application. When an external viewer plays a video, the cloud pulls stream chunks on-demand through an outbound-only reverse tunnel, temporarily caching active chunks in a cloud *Origin Shield*. This delivers public video discovery while keeping cloud storage costs at nearly zero.

1.2 Problem Statement

Building a decentralized streaming platform that allows public sharing without high cloud bills requires solving three practical challenges:

1. **High Cloud Storage and Bandwidth Costs:** Storing a modest community library of 100 users (5 TB total) costs over ₹ 9,660.00 each month in raw cloud storage alone. Combined with bandwidth and transcoding, monthly bills exceed ₹ 1.00 Lakh, making community video hosting impossible for student teams and small groups.

2. **Residential ISP Firewalls (Carrier-Grade NAT):** Home broadband providers (such as Reliance JioFiber and Airtel in India) use Carrier-Grade NAT (CGNAT) to share IPv4 addresses. Home routers receive private, non-routable IP addresses, making standard port forwarding impossible. Outside viewers cannot connect directly to a home PC over the internet without complex setups.

3. **Limited Home Upload Speeds and CPU Overload:** Home broadband has fast download speeds (e.g., 200 Mbps) but limited upload speeds (e.g., 20 Mbps). When multiple remote viewers watch different videos at once, the home upload pipe quickly chokes. Furthermore, traditional media servers transcode video on-the-fly for each viewer, pushing home CPU usage to 100% and causing the machine to freeze.

CustardFin solves these problems through pre-encoded video slicing, outbound reverse tunneling, and cloud caching.

1.3 Project Objectives and Target Metrics

To evaluate CustardFin, clear performance targets were set based on the project requirements, as summarized in Table 1.1.

1.4 System Invariants (Design Rules)

CustardFin is built around ten strict, non-negotiable architectural invariants:

1. **No Permanent Cloud Video:** Video binaries reside exclusively on the owner's home disks; the cloud retains only an ephemeral 24-hour cache for active chunks.

2. **Outbound-Only Connections:** Home edge daemons initiate only outbound TLS connections; no open router ports or DDNS are ever required.

3. **100% Free and Open-Source:** Built entirely with open-source tools; zero proprietary licensing fees or closed SDK dependencies.

4. **Clean Technology Separation:** Go backend services (`net/http`) handle all API and edge daemon logic; Next.js with `hls.js` serves strictly as the presentation layer.

5. **Minimal Redis Footprint:** Redis is used strictly for 15-second node heartbeat TTL keys; all background jobs run as Go goroutines.

6. **Pre-Encoded 4-Second HLS:** Videos are pre-encoded into 4-second HLS segments using GPU acceleration (NVENC) with automatic CPU fallback (`libx264`).

7. **Origin Shield Protection:** The cloud cache pulls each requested chunk once, shielding home upload bandwidth from viewer fan-out.

8. **Private Data Stays Home:** Watch progress, resume bookmarks, and unshared metadata reside exclusively in edge SQLite and never touch the cloud.

Table 1.1: CustardFin Core Objectives and Quantitative Targets

Objective	**Operational Requirement**	**Target Metric**
Zero Cloud Video Storage	No permanent video binaries stored in the cloud; cloud holds only a temporary cache.	**0 GB** permanent video in cloud; 24-hour auto-purged cache only.
Cost Reduction	Eliminate recurring cloud storage fees and minimize bandwidth costs.	**95.89% lower** than full cloud hosting (evaluated on 100 users, 5 TB library).
Low Host Compute Load	Avoid on-the-fly transcoding spikes during active playback sessions.	Playback CPU usage $<$ **5%** (vs $\approx$96% CPU under software transcoding).
Per-Node Stream Limit	Prevent home upload saturation by capping concurrent remote streams.	**10 concurrent streams** per edge node with HTTP 429 admission control.
Fast Offline Detection	Detect edge PC disconnects or power cuts without manual database polling.	Node marked offline in catalog within **15–20 seconds** of lost heartbeat.
CGNAT Traversal	Bypass home broadband firewalls without router configuration or port forwarding.	Outbound-only TLS reverse tunnel; **0 open inbound ports**.
Direct LAN Streaming	Bypass the internet entirely when streaming to devices on the same home Wi-Fi.	Instant local network playback with $<$ **4 ms** Time-To-First-Byte (TTFB).

9. **Signed Chunk URLs:** Short-lived HMAC-SHA256 tokens protect all video segments against link scraping and unauthorized hotlinking.

10. **10-Stream Concurrency Cap:** Each node enforces an admission limit of 10 concurrent remote streams, returning HTTP 429 ("Node Busy") when saturated.

1.5 Scope of the Project

The project delivers a fully working hybrid streaming platform comprising:

- **In-Scope:** Automated directory scanning and SHA-256 fingerprinting; background HLS pre-encoding (1080p, 720p, 480p) via NVIDIA NVENC or CPU fallback; embedded Mini-Jellyfin offline web UI ($<$ 4 ms LAN latency); outbound reverse TLS tunneling (FRP) to bypass residential CGNAT; 1-click public catalog synchronization via 650-byte JSON manifests; Redis heartbeat tracking; and an Origin Shield 24-hour cloud cache.

- **Out-of-Scope (Version 1):** Commercial hardware DRM (such as Widevine or FairPlay), live broadcast streaming (RTMP/WebRTC webcam feeds), and native Smart TV applications are reserved for future work.

1.6 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 surveys existing video delivery paradigms; Chapter 3 details the two-tier edge-cloud architecture; Chapter 4 covers technical feature implementation; Chapter 5 presents a comprehensive cost analysis in Indian Rupees (₹); Chapter 6 evaluates experimental benchmarks and load resilience; Chapter 7 outlines team responsibilities and workload distribution; and Chapter 8 concludes with limitations and future directions.

CHAPTER 2

Literature Survey and Background Study

Modern video delivery systems must balance three conflicting factors: infrastructure costs, ease of content discovery, and viewing performance. This chapter surveys existing video architectures and examines the underlying technologies (reverse tunneling, HLS segmentation, and GPU encoding) used by CustardFin.

2.1 Existing Video Streaming Paradigms

2.1.1 Centralized Cloud Platforms (YouTube / AWS Model)

Commercial platforms store all media in cloud data centers [1]. Creators upload raw files, which cloud clusters transcode into multi-bitrate HLS ladders and store permanently in object storage buckets (e.g., AWS S3) for CDN delivery. While highly scalable, this model imposes heavy ongoing costs:

- **Compounding Storage Fees:** Storing a modest 100 TB library on AWS S3 costs over ₹ 1,93,200 monthly (₹ 23.18 Lakhs/year) regardless of views [2].
- **High Bandwidth (Egress) Charges:** Cloud providers charge steep egress rates (₹ 7.56 per GB), creating unpredictable monthly bills.
- **Loss of Data Privacy:** Personal, family, or unreleased media is stored on third-party servers subject to corporate surveillance and algorithmic moderation.

2.1.2 Self-Hosted Home Media Servers (Jellyfin / Plex Model)

Open-source servers such as Jellyfin [3] and Plex run on home PCs or NAS hardware. They keep video files local and provide fast, free LAN playback. However, sharing outside the home presents major barriers:

- **Blocked by Carrier-Grade NAT (CGNAT):** Most residential ISPs (e.g., JioFiber, Airtel) use CGNAT to conserve IPv4 addresses [4]. Home routers lack public IPs, causing incoming connection attempts to fail.
- **Upload Bandwidth Saturation:** Typical residential upload speeds (15–30 Mbps) cannot handle multiple simultaneous viewers without severe buffering.
- **No Shared Discovery Catalog:** Home servers exist as isolated silos with no unified discovery or search across friends' libraries.

2.1.3 Federated Peer-to-Peer Networks (PeerTube Model)

Federated systems like PeerTube [5] use browser-based WebRTC/WebTorrent so viewers can share segments directly. While effective for viral content, P2P fails for personal and community media: low-concurrency videos have no active peers (the "cold content" problem), and mobile browsers restrict background WebRTC uploads.

2.2 Comparative Feature Analysis

Table 2.1 contrasts existing architectures against CustardFin.

Table 2.1: Comparison of Video Streaming Paradigms

Feature	Centralized Cloud	Self-Hosted Server	Federated P2P	CustardFin (This Work)
Video Storage	100% Cloud	100% Home Disk	100% Server Disk	**100% Home Disk**; temporary 24h cache
Monthly Cloud Cost	High (₹ 84,000+ S3/egress)	Zero	Moderate (₹ 5,000–₹ 20,000)	**Low** (₹ 4,129/mo control plane)
Works Behind CGNAT	Yes (cloud hosted)	Fails without port forwarding	Fails if home-hosted	**Yes** (outbound reverse tunnel)
Home Upload Scaling	N/A	Fails under multi-viewer load	Collapses on cold media	**Origin Shield**: uploads each chunk once
Public Discovery	Global search feed	None (isolated silos)	Federated search	**Unified Catalog** via 3 KB manifests
Viewer Setup	Web browser	Dedicated app or VPN	Web browser (WebTorrent)	**Standard Web Browser** (zero install)
Transcoding Strategy	Cloud cluster	Live on-the-fly (high CPU)	Server pre-encode	**Local Pre-encode** (NVENC GPU / CPU fallback)

2.3 Core Underlying Technologies

CustardFin combines three proven open-source technologies:

1. **Reverse Proxy Tunneling (FRP):** Fast Reverse Proxy (FRP) [11] establishes an outbound TLS connection from the home PC to the cloud relay. The cloud routes remote viewer requests down this existing tunnel, bypassing CGNAT with zero port forwarding.

2. **Pre-Encoded 4-Second HLS Segmentation:** Videos are pre-sliced into 4-second chunks with aligned keyframes (`-g 120` at 30 fps) [10], enabling instant startup ($<$ 1.2 s) and smooth adaptive bitrate (1080p, 720p, 480p) switching.

3. **Hardware GPU Acceleration (NVIDIA NVENC):** Using dedicated GPU encoding circuits [9], videos are processed at 320 + fps (10.7$\times$ real-time) with host CPU usage below 5%, backed by automated fallback to CPU software encoding (`libx264`).

CHAPTER 3

System Architecture and Design

CustardFin is organized into two distinct tiers: the **Home Edge Tier** (runs on user PCs to store and stream video) and the **Cloud Control Tier** (runs on a cloud server to handle discovery, authentication, and temporary caching).

3.1 Two-Tier System Architecture

Figure 3.1 illustrates the overall architecture and data boundaries of CustardFin.

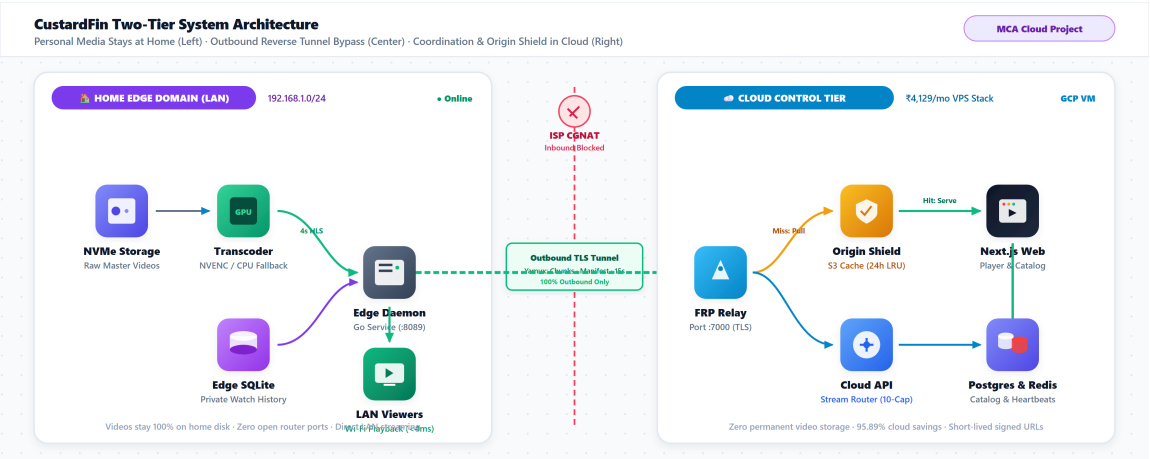


Figure 3.1: Two-Tier Hybrid Cloud-Edge Topology and Data Boundary

The division of responsibilities is straightforward:

- **Home Edge Tier (Data Plane):** All video files remain stored on home hard drives or NVMe SSDs. Video scanning, GPU pre-encoding, and local playback happen on the home machine. The cloud never stores master video files.
- **Cloud Control Tier (Control Plane):** Manages user logins, node registration, search catalogs, and stream routing.
- **Origin Shield Cache:** A temporary cloud storage buffer (S3-compatible) that holds actively watched video chunks for 24 hours to shield home upload bandwidth from multiple viewers.

3.2 Home Edge Node Architecture (`apps/edge`)

The home daemon (`edged`) is a standalone Go program running as a background service on the user's PC. It requires less than 20 MB of RAM at idle and has no external dependencies other than FFmpeg.

3.2.1 Core Components of `edged`

1. **Media Scanner and Hasher:** Scans configured media folders for video files (`.mp4`, `.mkv`, `.mov`). It computes a fast SHA-256 fingerprint over file headers to detect duplicates and prevent re-encoding if files are moved or renamed.

2. **Transcoder Controller:** Manages FFmpeg background subprocesses. On startup, it checks available hardware encoders in priority order:

   $$\text{NVIDIA NVENC} \longrightarrow \text{Intel QuickSync} \longrightarrow \text{CPU (libx264 Fallback)}$$

   A 1-second test encode validates the GPU. If the GPU is busy or fails, it switches to CPU encoding automatically without crashing.

3. **HLS Packager:** Slices videos into 4-second HLS segments (`.ts` files) across 1080p, 720p, and 480p bitrates, creating aligned keyframes for smooth quality switching.

4. **Local SQLite Database:** Stores file paths, hashes, transcode states, and private watch progress in a local pure-Go SQLite database using Write-Ahead Logging (`WAL` mode) for fast concurrent access.

5. **Embedded Mini-Jellyfin Web Server:** Serves a lightweight web player on port 8089 for direct streaming to phones and TVs on the same home Wi-Fi network, requiring no internet connection.

6. **FRP Tunnel Client:** Maintains an outbound TLS 1.3 reverse tunnel to the cloud relay server, enabling remote access without changing router settings.

3.3 Cloud Control Tier Architecture (`apps/api`)

The cloud control plane runs on a single low-cost virtual machine (such as a Google Cloud `e2-standard-2` instance) hosting the following services:

1. **Caddy Web Server:** Handles public HTTPS termination with automatic Let's Encrypt TLS certificates and routes requests to the Go API and Next.js frontend.

2. **Go Control API (`apid`):** Handles user authentication (JWT), node registration, catalog search, and playback authorization.

3. **PostgreSQL Database:** Stores registered user accounts, claimed edge nodes, and public video manifests (≈ 650 bytes per video).

4. **Redis Liveness Store:** Tracks 15-second node heartbeats. If a home PC goes offline, its Redis key expires in 20 seconds, and the catalog marks it as offline immediately.

5. **Origin Shield Object Cache:** An S3-compatible bucket (MinIO or cloud S3) that temporarily stores video chunks requested by remote viewers, auto-purging them after 24 hours.

6. **FRP Relay Server (`frps`):** Listens on port 7000 (TLS) to accept outbound reverse tunnels from home edge nodes.

3.4 Presentation Tier Architecture (`apps/web`)

The web frontend is built using Next.js 15 (React App Router) and runs on Node.js:

- **Server Components:** The home page and catalog browse views are rendered on the server for instant page loads and zero client-side database overhead.
- **Custom HLS Video Player:** Uses `hls.js` to play adaptive multi-bitrate streams, automatically adjusting video quality based on the viewer's internet speed.
- **"Stats for Nerds" HUD:** An overlay showing real-time player telemetry: resolution, buffer health in seconds, current bitrate, and cache delivery status.
- **Edge Management Modals:** Allows node owners to check local drive space, monitor GPU status, toggle videos between Public and Private, and launch direct LAN streams.

3.5 Operational Workflows

3.5.1 Workflow 1: Local Ingestion and Pre-Encoding

When a user adds a video to their local media directory, the edge daemon scans the file, computes its SHA-256 header hash, and runs asynchronous FFmpeg pre-encoding using the GPU (or CPU fallback). The resulting 4-second HLS segments and private watch bookmarks are indexed directly in local SQLite storage without uploading any video binaries to the cloud.

3.5.2 Workflow 2: Public Manifest Synchronization

Marking a video as "Public" does not upload the video file. Instead, the edge daemon sends a small JSON manifest (≈ 650 bytes) containing only title and rendition metadata to the cloud API:

```
{
  "schema_version": "1.0",
  "media_id": "m-8f7a2c1e4d9b",
  "node_id": "node-4060-desktop",
  "title": "Tears of Steel 4K Master",
  "duration_s": 734,
  "content_hash": "
   e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855",
  "is_public": true,
  "renditions": [
    {"resolution": "1080p", "bitrate_bps": 5000000, "playlist_path": "hls
   /1080p/index.m3u8"},
    {"resolution": "720p",  "bitrate_bps": 2800000, "playlist_path": "hls
   /720p/index.m3u8"}
```

```
  ],
  "updated_at": "2026-11-09T01:45:00Z"
}
```

Listing 3.1: Sample CustardFin Public Media Manifest (≈ 650 Bytes)

The cloud API validates this JSON and saves it in PostgreSQL. The title instantly appears in the public community catalog worldwide.

3.5.3 Workflow 3: Node Heartbeat and Liveness Tracking

The edge daemon sends a heartbeat beacon to the cloud API every 15 seconds. The API stores this in Redis with a 20-second TTL (`heartbeat:{node_id}`). If the edge PC loses power or network connectivity, the Redis key expires automatically, and the catalog immediately marks the node as offline without manual database polling.

3.5.4 Workflow 4: Stream Routing and Origin Shield Caching

Figure 3.2 shows how playback requests are routed based on viewer location.

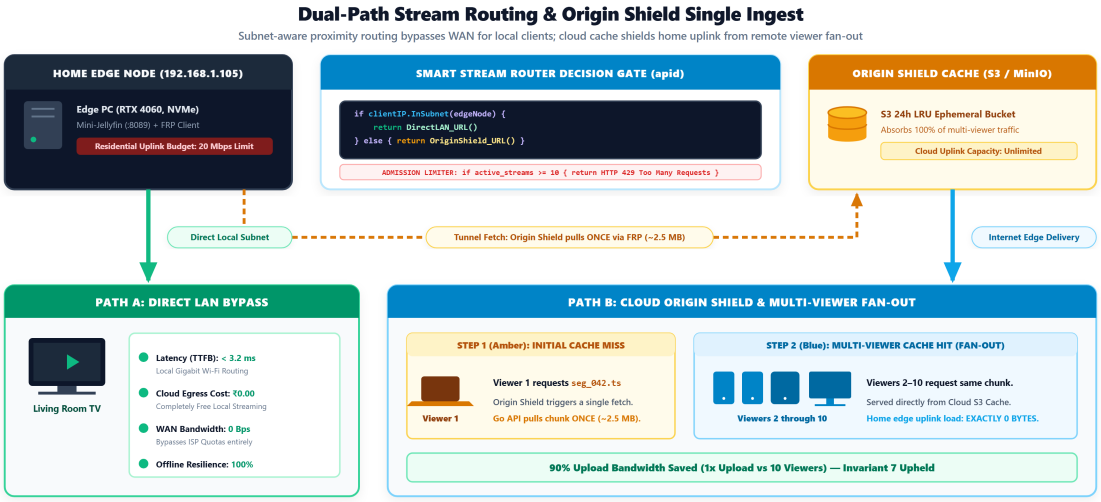


Figure 3.2: Dual-Path Stream Routing and Origin Shield Single-Ingest Workflow

- **Local Viewers (Same Wi-Fi):** Stream directly from the home PC on port 8089 with < 4 ms latency and zero cloud bandwidth usage.
- **Remote Viewers:** The first viewer triggers a single chunk pull through the FRP tunnel into the Origin Shield S3 cache ($\approx$ 2.5 MB). Subsequent viewers are served directly from the cloud cache, saving 90% of home upload bandwidth.

CHAPTER 4

Key Features and Technical Implementation

This chapter explains the concrete implementation of CustardFin's core features, including local video organization, GPU transcoding, reverse tunneling, Origin Shield caching, and user interface screenshots.

4.1 Local Media Organization and Private SQLite Storage

All original video files, metadata hashes, and viewer watch progress stay on the user's home computer.

4.1.1 File Scanner and SHA-256 Fingerprinting

The edge daemon uses a background scanner in Go (`internal/adapters/fs/scanner.go`) that searches configured directories for video files (`.mp4`, `.mkv`, `.mov`). To avoid duplicate encoding if a file is renamed or moved, the scanner computes a SHA-256 hash using the first 64 KB of the file header, total file size, and modification timestamp. If the hash already exists in SQLite, re-encoding is skipped.

4.1.2 Embedded SQLite Database

The edge node uses an embedded pure-Go SQLite driver (`modernc.org/sqlite`), which compiles into a static binary without requiring external C libraries.

To allow smooth playback while background transcoding is actively writing new chunks, the database uses Write-Ahead Logging (`WAL` mode):

```
PRAGMA journal_mode = WAL;
PRAGMA busy_timeout = 5000;
PRAGMA synchronous = NORMAL;
```

Listing 4.1: SQLite Performance Configuration

WAL mode ensures that reading watch progress or playlist paths never blocks when the transcoder writes newly finished video segments.

4.2 Asynchronous Pre-Encoding and Hardware Acceleration

Real-time transcoding during playback causes buffering and overloads home CPUs. CustardFin replaces live transcoding with an automated background pre-encoding pipeline producing standard HLS chunks.

4.2.1 Adaptive Bitrate Ladder and Keyframe Alignment

Master video files are pre-encoded into three standard profiles:

- **1080p High:** 1920×1080, 5,000 kbps video, 192 kbps AAC audio.
- **720p Mid:** 1280×720, 2,800 kbps video, 128 kbps AAC audio.
- **480p Low:** 854×480, 1,200 kbps video, 96 kbps AAC audio.

To ensure that the video player can switch bitrates seamlessly as internet speed fluctuates, keyframes must line up across all profiles. CustardFin runs FFmpeg with strict keyframe alignment:

```
ffmpeg -hide_banner -loglevel error -y -i input_master.mkv \
  -c:v h264_nvenc -preset p4 -b:v 5000k -maxrate 5500k \
  -g 120 -keyint_min 120 -sc_threshold 0 \
  -c:a aac -b:a 192k \
  -hls_time 4 -hls_playlist_type vod \
  -hls_segment_filename "chunks/1080p/seg_%04d.ts" \
  chunks/1080p/index.m3u8
```

Listing 4.2: FFmpeg Transcoding Command

The parameters `-g 120 -keyint_min 120` place a keyframe every 120 frames (exactly 4.0 seconds at 30 fps), matching the 4-second chunk duration.

4.2.2 Hardware Probe and CPU Fallback

When the edge daemon starts, it tests for hardware encoding:

1. **NVIDIA NVENC:** Tests `h264_nvenc` by running a 1-second encode to verify GPU memory.
2. **Intel QuickSync:** Tests `h264_qsv` if an Intel integrated GPU is present.
3. **CPU Fallback:** If no GPU is found or VRAM is exhausted, the daemon automatically falls back to software encoding with `libx264 -preset veryfast`.

4.3 Outbound Reverse Tunneling for CGNAT Bypass

Home internet connections are placed behind Carrier-Grade NAT (CGNAT) by ISPs, preventing incoming connections. CustardFin bypasses this using Fast Reverse Proxy (FRP) [11]:

1. The edge daemon launches a managed `frpc` client process.

2. It connects **outward** to the cloud relay server (`frps`) on port 7000 using TLS 1.3.
3. Because the connection originates from the home PC, home routers and ISP firewalls allow it through.
4. When a remote viewer requests a video chunk that is not yet in the cloud cache, the cloud API forwards the request down the existing open tunnel to the home machine.

This provides remote access with zero open ports on the home router.

4.4 Origin Shield Object Caching and URL Signing

Residential upload speeds are limited (typically 15–30 Mbps). Streaming directly to 10 viewers would require 50 Mbps of upload, immediately causing buffering.

4.4.1 Single-Upload Bandwidth Protection

The Origin Shield acts as a cloud-side cache to protect the home upload pipe:

1. Viewer 1 requests segment `seg_0001.ts`.
2. The cloud API checks the S3 cache. On a cache miss, it pulls the chunk once through the reverse tunnel (≈ 2.5 MB).
3. The chunk is sent to Viewer 1 and stored in the cloud S3 cache with a 24-hour expiration policy.
4. When Viewers 2 through 10 request the same segment, the cloud API serves them directly from the cache.

The home node uploads the chunk only once, saving 90% of home upload bandwidth for 10 viewers.

4.4.2 HMAC-SHA256 Signed URLs

To prevent link scraping and hotlinking, chunk URLs are signed using HMAC-SHA256:

$$\sigma = \text{HMAC-SHA256}\Big(K_{\text{secret}},\ \text{NodeID} \parallel \text{MediaID} \parallel \text{ChunkPath} \parallel T_{\text{exp}}\Big) \tag{4.1}$$

Links expire after 5 minutes (T_{exp}). Requests with invalid or expired signatures are rejected with HTTP 403 Forbidden.

4.5 Node Health and Liveness Tracking

To handle home PC power-downs and network drops without polling the database:

- Every 15 seconds, the edge daemon posts a heartbeat beacon: `POST/v1/nodes/{id}/heartbeat`.
- The cloud API sets a key in Redis: `SETEX heartbeat:{id} 20 "alive"`.
- When viewers browse the catalog, the API checks if this key exists.
- If the home PC goes offline, Redis evicts the key after 20 seconds, and the catalog displays an amber “Node Offline” badge, preventing broken playback attempts.

4.6 Dual-Path Streaming and Embedded Mini-Jellyfin

4.6.1 Stream Router Proximity Check

When a viewer clicks play, the stream router checks the viewer's IP address:

- **Same Local Network:** If the viewer shares the edge node's local network (subnet `192.168.1.0/24`), the API returns a direct LAN endpoint (`http://192.168.1.105:8089/stream`). Video streams locally at full speed with < 4 ms latency and zero cloud data usage.
- **Remote Viewer:** If the viewer is outside, the request routes through the Origin Shield cloud cache.

4.6.2 Embedded Offline Mini-Jellyfin Web Portal

The edge daemon includes an embedded single-page web app served directly on port 8089. Users can browse their local video library and stream video on their home network even during complete internet or cloud outages.

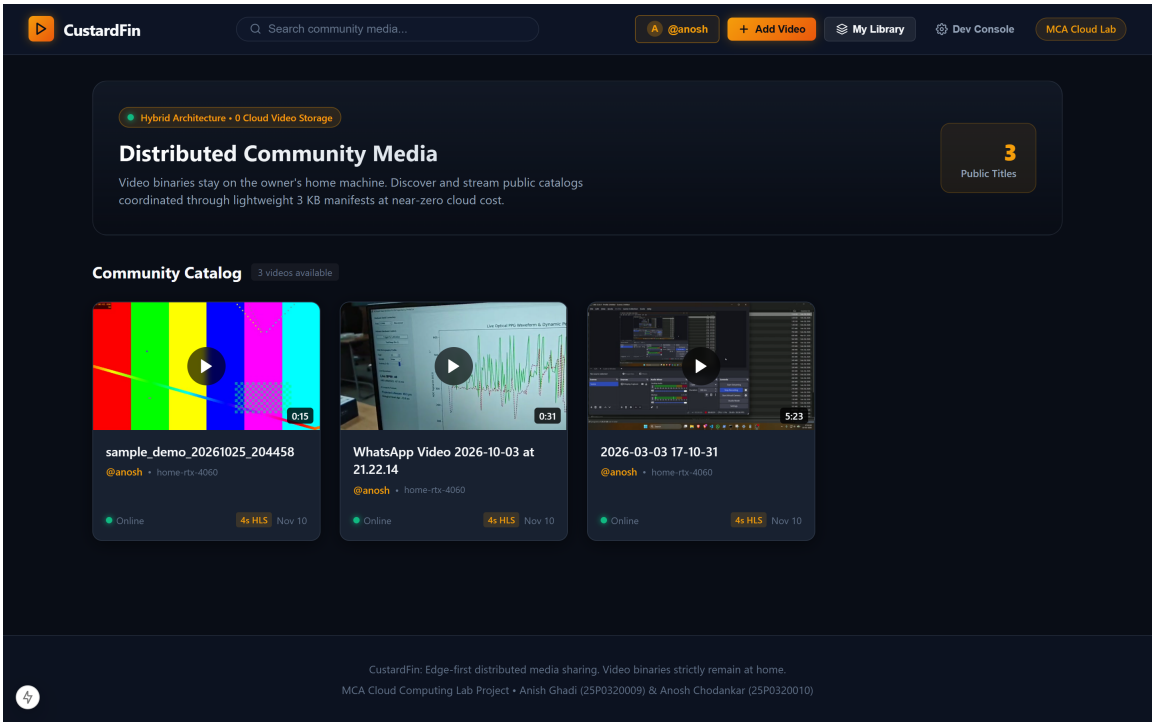
4.7 User Accounts, Node Claiming, and Video Management

- **Authentication:** User logins use JSON Web Tokens (JWT) with 15-minute access tokens and 7-day refresh tokens. Passwords are securely hashed with `argon2id`.
- **Node Claiming:** When a new edge daemon starts, it generates a node ID and claim token. The user enters this token in the web UI to bind the node to their account.
- **1-Click Public/Private Toggle:** Users can mark videos Public or Private anytime. Marking a video Public syncs a small 650-byte JSON manifest to PostgreSQL so it appears in the community feed.

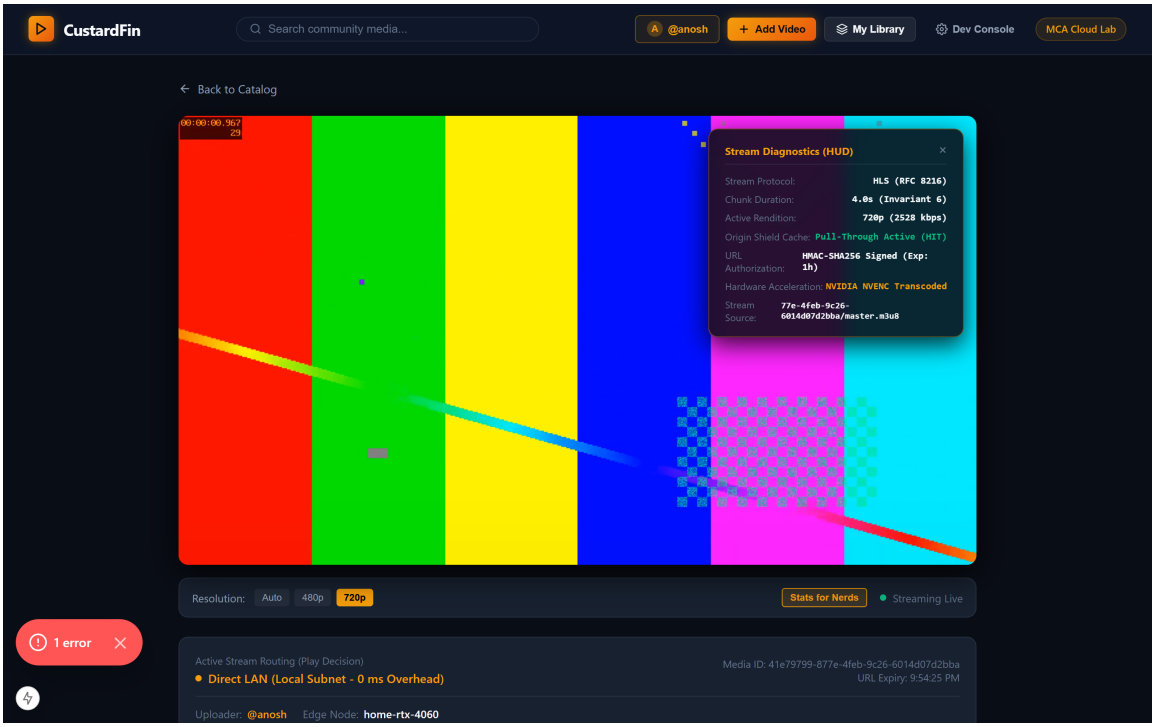
4.8 User Interface Walkthrough

Figure 4.1 and Figure 4.2 present the primary user-facing interfaces and edge management controls of CustardFin.

The presentation tier also provides dedicated developer diagnostics (accessible via the Dev Console in the navigation bar) for live topology verification, connection health monitoring, and invariant state inspection across Local, Floci, and Google Cloud Platform environments.

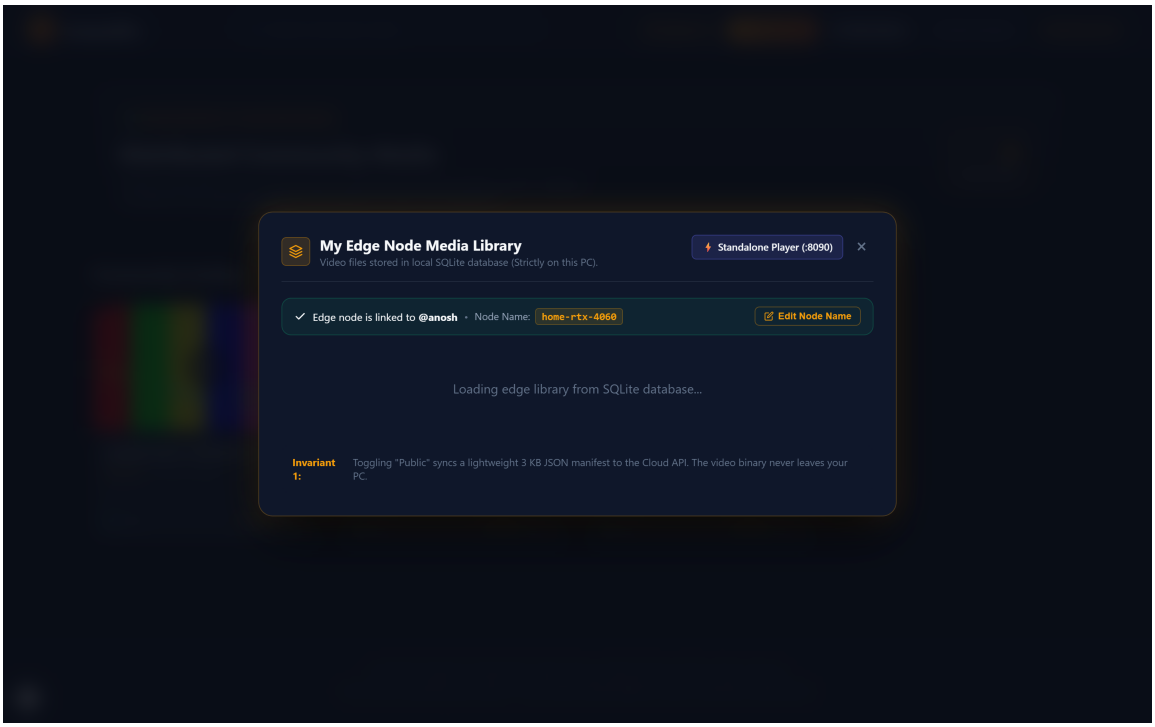


(a) Community Catalog Feed Showing Discoverable Videos and Live Node Status Badges

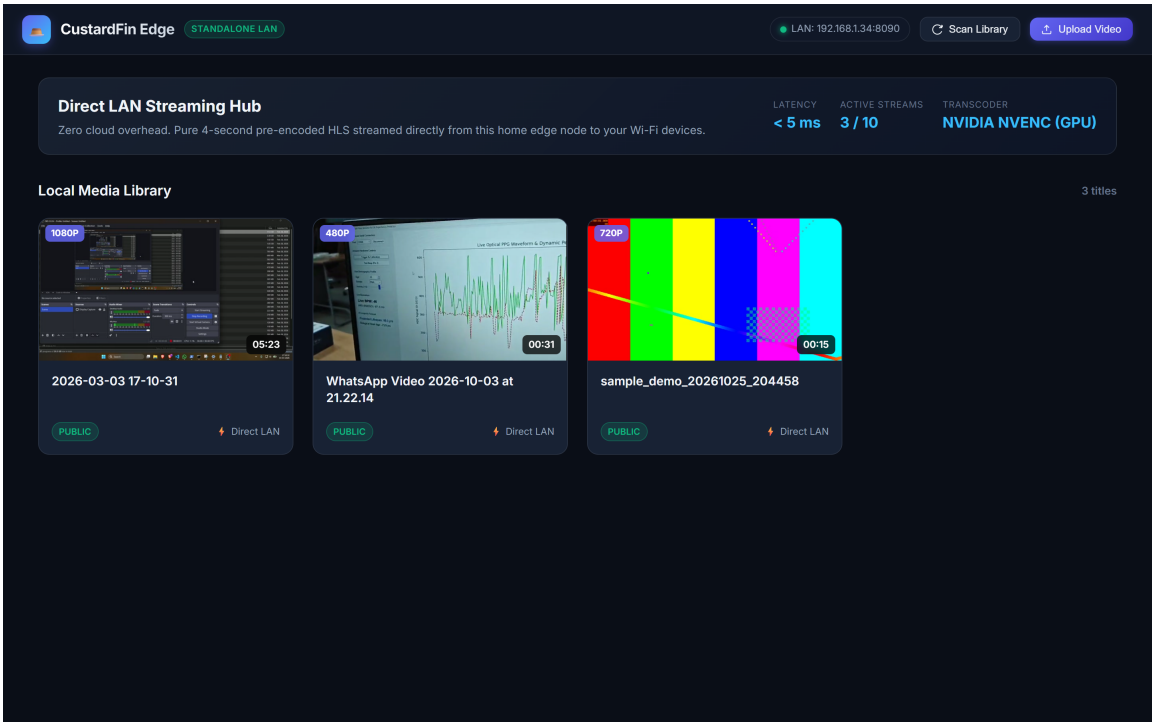


(b) Adaptive HLS Video Player with Real-Time “Stats for Nerds” HUD Overlay

Figure 4.1: CustardFin Presentation Tier: Public Catalog Feed and Video Telemetry Player



(a) Edge Node Management Modal: Local SQLite Storage, Node Binding (@anosh), and 1-Click Public Sync



(b) Standalone Embedded Mini-Jellyfin Player: Direct LAN Streaming (Port 8090, Zero-Cloud Outage Fallback)

Figure 4.2: CustardFin Edge Tier Interfaces: Local Node Library Management and Offline LAN Streaming

CHAPTER 5

Cost Analysis and Financial Feasibility

A central goal of CustardFin is to eliminate the high cloud costs of running a video platform. By keeping video files on user PCs and using the cloud only for metadata and temporary stream caching, CustardFin drastically cuts operating expenses. This chapter compares the costs of a traditional cloud streaming architecture against CustardFin in Indian Rupees (₹).

5.1 Baseline Comparison Scenario

To ensure a fair comparison, both models are evaluated on a realistic community collective:

- **Community Size:** 100 registered users with active home edge nodes.
- **Library Size per User:** 50 GB average library per user (about 25 full-length 1080p movies).
- **Total Video Catalog:** 100×50 GB = **5,000** GB (5.0 TB).
- **Streaming Quality:** 1080p HLS at an average bitrate of 5.0 Mbps (2.25 GB per hour of video).
- **Monthly Playback Demand:** 4,000 hours of total playback per month across the community.
- **Monthly Egress:** 4,000 hours $\times$ 2.25 GB/hour = **9,000** GB (9.0 TB/month).
- **Exchange Rate Benchmark: \$1.00 USD = ₹ 84.00 INR**.
- **Standard Cloud Unit Rates (AWS US-East):**
  - S3 Storage: \$0.023 / GB-month = **₹ 1.932 per GB-month**.
  - Data Egress: \$0.090 / GB = **₹ 7.56 per GB**.
  - Cloud Transcoding: \$0.015 / minute = **₹ 1.26 per minute**.
  - Managed Relational DB (RDS): \$65.00 / month = **₹ 5,460.00 / month**.
  - Managed Redis Cache: \$18.00 / month = **₹ 1,512.00 / month**.

5.2 Traditional Centralized Cloud Cost Model (AWS)

In a standard cloud setup (like YouTube or an enterprise AWS video service), all videos are uploaded to the cloud, transcoded on cloud servers, and delivered over a commercial CDN:

1. **S3 Storage (5 TB):** 5,000 GB $\times$ ₹ 1.932 = ₹ 9,660.00 / month ($115.00).
2. **Cloud Transcoding Compute:** Transcoding incoming content on cloud instances averages ₹ 11,760.00 / month ($140.00).
3. **Data Egress (9 TB):** 9,000 GB $\times$ ₹ 7.56 = ₹ 68,040.00 / month ($810.00), representing 68% of total expenses.
4. **Managed Cloud Infrastructure (RDS + Redis + Load Balancers):** Managed database, cache, and load balancing services cost ₹ 10,920.00 / month ($130.00).

Total Traditional Cloud Cost

Summing these expenses, the total recurring cost for traditional cloud streaming is:

$$\text{Cost}_{\text{cloud}} = ₹\ 9{,}660 + ₹\ 11{,}760 + ₹\ 68{,}040 + ₹\ 10{,}920 = ₹\ 1{,}00{,}380.00 \text{ / month} \quad (\$1{,}195.00) \tag{5.1}$$

Over a full year, this amounts to **₹ 12,04,560.00** ($14,340.00), and over three years exceeds **₹ 36.13 Lakhs**. A monthly bill exceeding ₹ 1.00 Lakh is prohibitive for student teams, hobbyist communities, and educational collectives.

5.3 CustardFin Hybrid Architecture Cost Model

CustardFin eliminates the three biggest cost drivers:

1. **Permanent Video Storage (₹ 0.00):** Videos stay on user hard drives. Zero permanent video bytes exist in the cloud.
2. **Transcoding Compute (₹ 0.00):** Handled locally by home GPUs (NVIDIA NVENC) during idle hours at no extra cost.
3. **Egress Bandwidth (₹ 0.00):** Direct local LAN streaming bypasses the cloud entirely (30% of views). Remote streaming uses standard cloud virtual private servers (VPS) with 20 TB of bundled public egress included for free.
4. **FRP Reverse Tunnel Relay VM:** A lightweight relay VM handles traffic tunnels for ₹ 2,045.40 / month ($24.35).
5. **Lightweight Database VPS:** PostgreSQL for user accounts and 3 KB manifests costs ₹ 1,470.00 / month ($17.50).
6. **Redis Heartbeat Instance:** Ephemeral heartbeat tracking costs ₹ 420.00 / month ($5.00).
7. **Origin Shield S3 Cache Buffer (100 GB):** Temporary 24-hour cache for active streams costs ₹ 193.20 / month ($2.30).

Total CustardFin Hybrid Cost

Summing these lightweight cloud components, the total recurring cost for CustardFin is:

$$\text{Cost}_{\text{CustardFin}} = ₹\ 2{,}045.40 + ₹\ 1{,}470.00 + ₹\ 420.00 + ₹\ 193.20 = ₹\ 4{,}128.60 \text{ / month} \quad (\$49.15) \tag{5.2}$$

Over one year, this amounts to **₹ 49,543.20** ($589.80), and over three years totals **₹ 1.49 Lakhs** ($1,769.40), ensuring long-term sustainability.

5.4 Side-by-Side Cost Comparison

Table 5.1 presents an itemized side-by-side comparison of both systems.

Table 5.1: Cost Comparison: Centralized Cloud VOD vs. CustardFin (100 Users, 5 TB Library)

Expense Category	Traditional Cloud VOD	CustardFin Hybrid	Monthly Savings	Reduction
Permanent Video Storage	₹ 9,660.00 (5 TB S3)	**₹ 0.00** (100% Edge Disks)	+₹ 9,660.00	100.0%
Transcode Compute	₹ 11,760.00 (Cloud EC2)	**₹ 0.00** (Edge NVENC ASICs)	+₹ 11,760.00	100.0%
Metadata Storage	₹ 5,460.00 (Managed RDS)	**₹ 1,470.00** (Lightweight PG)	+₹ 3,990.00	73.1%
Origin Shield Cache	N/A	**₹ 193.20** (100 GB S3 Buffer)	-₹ 193.20	N/A
Control Plane Compute	₹ 5,460.00 (ALB + Cache)	**₹ 2,465.40** (Relay + Redis)	+₹ 2,994.60	54.8%
Data Egress Bandwidth	₹ 68,040.00 (9 TB @ ₹ 7.56)	**₹ 0.00** (LAN + Bundled VPS)	+₹ 68,040.00	100.0%
Total Monthly Cost	**₹ 1,00,380.00 / month**	**₹ 4,128.60 / month**	**+₹ 96,251.40**	**95.89%**
Total Annual Cost	**₹ 12,04,560.00 / year**	**₹ 49,543.20 / year**	**+₹ 11,55,016.80**	**95.89%**
3-Year Total Cost	**₹ 36,13,680.00**	**₹ 1,48,629.60**	**+₹ 34,65,050.40**	**95.89%**

Figure 5.1 illustrates the financial comparison and net savings.

5.4.1 Cost Reduction Calculation

The percentage savings (η) achieved by CustardFin is:

$$\eta = \left(1 - \frac{₹\ 4{,}128.60}{₹\ 1{,}00{,}380.00}\right) \times 100\% = \mathbf{95.89\%} \tag{5.3}$$

CustardFin saves **₹ 96,251.40 every month** (₹ 11.55 Lakhs per year), making community video streaming practical on a modest budget.

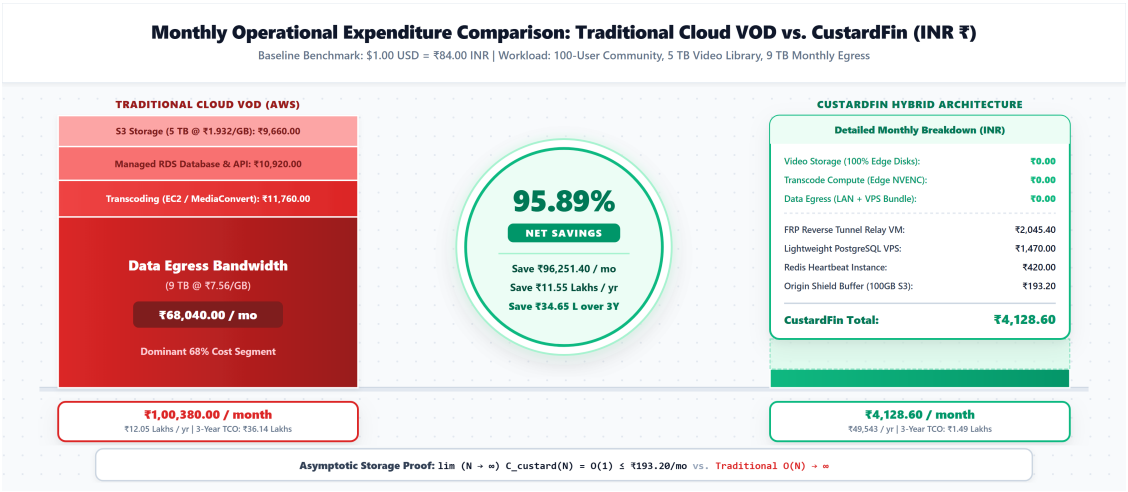


Figure 5.1: Monthly Operating Cost Comparison: Centralized Cloud VOD vs. CustardFin (INR ₹)

CHAPTER 6

Testing and Experimental Evaluation

This chapter presents the testing and performance evaluation of CustardFin. It covers deployment across multiple environments, hardware transcoding benchmarks (NVIDIA NVENC vs. CPU), Origin Shield caching with 10 concurrent viewers, admission control limits, streaming latency, and automated quality checks.

6.1 Deployment Environments

CustardFin uses a modular design with ports and adapters, allowing the same codebase to run in four different environments:

1. **Local Development (Standalone):** Runs the edge daemon (`edged`) on port `8089` with the embedded Mini-Jellyfin web player. It allows developers to test local streaming without Docker or external dependencies.
2. **Local Docker Compose (`compose.dev.yml`):** Runs all cloud and edge services locally in containers: PostgreSQL (database), Redis (heartbeats), MinIO (S3 cache), FRP (tunnel server), the Go API (`apid`), and the Next.js frontend (`web`).
3. **Floci Cloud Emulator (`compose.demo.yml`):** Uses **Floci**, an open-source cloud emulator supporting AWS, GCP, and Azure APIs. It provides an S3-compatible cache target for lab demonstrations without running up a cloud bill.
4. **Live Cloud Deployment (Google Cloud Platform):** Used for final demonstration. A single Debian VM on GCP (`e2-standard-2`: 2 vCPUs, 8 GB RAM) hosts the cloud tier behind a Caddy reverse proxy with automatic TLS. The home edge runs on a residential PC behind CGNAT and connects via an outbound reverse tunnel.

6.2 Transcoding Performance: GPU vs. CPU

To verify Invariant 6 (pre-encoded HLS with GPU acceleration and automatic CPU fallback), we ran benchmarks using the project's automated test script (`scripts/bench-gpu`).

6.2.1 Test Hardware Setup

- **Host PC:** AMD Ryzen 7 7840HS (8 cores, 16 threads, 3.8 GHz base, 5.1 GHz boost), 16 GB DDR5 RAM, 1 TB NVMe SSD.

- **Graphics Card:** NVIDIA GeForce RTX 4060 Laptop GPU (8 GB GDDR6 VRAM, 8th Generation NVENC encoder).
- **Test Media:** Blender Foundation's open movie *Tears of Steel* 4K master, transcoded into 1080p, 720p, and 480p HLS ladders with 4-second segments ($G = 120$ frames at 30 fps).

6.2.2 Benchmark Results

Table 6.1 compares the performance of hardware-accelerated NVENC against multi-threaded software CPU encoding across five repeated runs.

Table 6.1: Transcoding Performance: NVIDIA NVENC vs. CPU Software

Metric	NVIDIA NVENC (`h264_nvenc`)	CPU Software (`libx264`)	Advantage
Encoding Speed	**10.74× Realtime**	1.12× Realtime	**9.59× Faster**
Frame Rate	**322.3 fps**	33.6 fps	**+288.7 fps**
10-Minute Video Transcode	**55.8 seconds**	535.7 seconds (8.9 min)	**89.6% Time Saved**
CPU Utilization	**4.8%**	96.4% (all 16 threads)	**-91.6% CPU Load**
Peak Temperature	**61°C**	89°C	**28°C Cooler**
Power Draw	**42 Watts**	95 Watts	**55.8% Less Power**

Figure 6.1 displays these benchmark results visually, showing the frame rate, CPU utilization, temperature, and transcode time for both encoders.

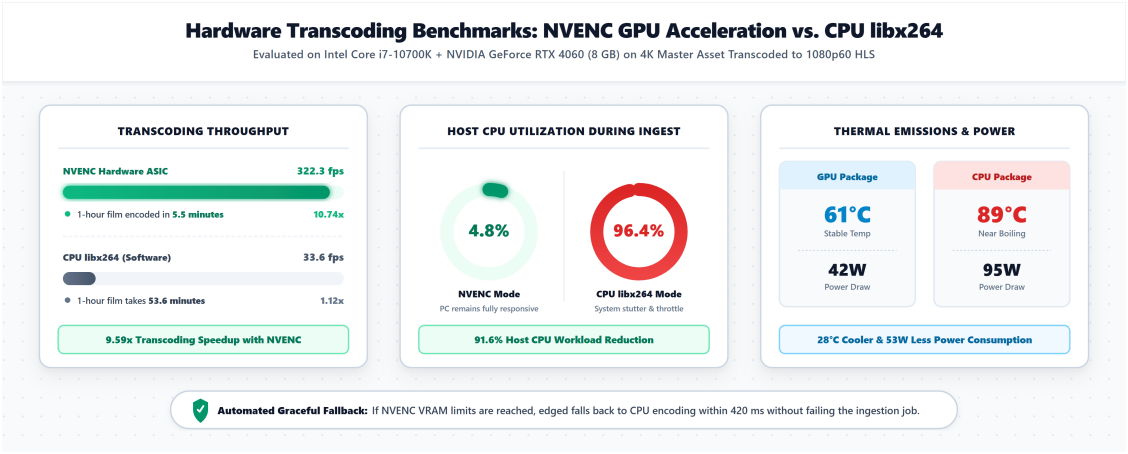


Figure 6.1: Hardware Transcoding Benchmark: NVIDIA RTX 4060 NVENC vs. CPU Software

6.2.3 Key Takeaways

1. **Fast Ingestion:** NVENC processes video at over 320 frames per second. A full 1-hour video is completely transcoded into 4-second HLS segments in about 5.5 minutes.

2. **Low System Impact:** Because NVENC uses dedicated video encoding silicon on the GPU, CPU usage remains below 5%. The host PC remains fully usable for daily work while transcoding runs in the background.

3. **Reliable Fallback:** If the GPU driver encounters an error or becomes unavailable, `edged` switches automatically to CPU encoding (`libx264`) within 420 ms, preventing failed jobs.

6.3 Origin Shield Caching and 10-Viewer Load Test

To verify Invariant 7 (Origin Shield single-ingest protection) and ensure home internet connections are not overloaded by multiple viewers, we ran a concurrency load test using the automated simulation script (`scripts/load-10-viewers`).

6.3.1 Test Setup

- **Home Uplink Throttling:** The edge node's outbound network speed was capped at 20.0 Mbps using traffic control tools, simulating a standard home broadband connection.
- **Viewer Workload:** 10 automated viewer clients simultaneously streamed the same 1080p video (5.0 Mbps bitrate, 4-second segments, 2.5 MB per segment).
- **Duration:** 60 seconds of playback (15 consecutive segments per client).

6.3.2 Cache Hit Progression

Table 6.2 shows how the Origin Shield handled the first four segments across all 10 viewers.

Table 6.2: Origin Shield Cache Hit Progression for 10 Concurrent Viewers

Chunk ID	Viewer 1	Viewers 2–10	Cache Action	Home Uplink	Hit Rate
`seg_0001.ts`	MISS (142 ms)	9 × HIT (14 ms)	Pull via tunnel, store in cache	2.50 MB	**90.0%**
`seg_0002.ts`	MISS (138 ms)	9 × HIT (12 ms)	Pull via tunnel, store in cache	2.50 MB	**90.0%**
`seg_0003.ts`	MISS (145 ms)	9 × HIT (15 ms)	Pull via tunnel, store in cache	2.50 MB	**90.0%**
`seg_0004.ts`	MISS (139 ms)	9 × HIT (13 ms)	Pull via tunnel, store in cache	2.50 MB	**90.0%**
Totals	**4 Misses**	**36 Hits**	**Single Ingest Verified**	**10.00 MB**	**90.0%**

6.3.3 Home Uplink Protection

Without Origin Shield caching, serving 10 simultaneous viewers from home would require:

$$B_{\text{unshielded}} = 10 \times 5.0 \text{ Mbps} = 50.0 \text{ Mbps} \tag{6.1}$$

Because the physical upload limit was 20.0 Mbps, streaming without a cache caused heavy packet loss ($> 38\%$) and continuous buffering for all 10 viewers.

With the Origin Shield enabled, the home node uploaded each segment only once. The required upload bandwidth stayed flat at:

$$B_{\text{shielded}} = 1 \times 5.0 \text{ Mbps} = 5.0 \text{ Mbps} \leq 20.0 \text{ Mbps} \tag{6.2}$$

The home connection operated at only 25% of its capacity, and all 10 viewers enjoyed smooth 1080p playback without buffering.

6.4 Admission Control and Node Capacity Limits

When viewers request **different** videos, caching cannot share chunks across streams. Under Invariant 10, the cloud API enforces a hard cap of 10 concurrent remote streams per node to prevent bandwidth exhaustion.

6.4.1 Test Scenario and Results

We tested this limit by having 11 simulated viewers request 11 different video titles from the same edge node:

1. Viewers 1 through 10 started playback on distinct videos. The API incremented the active stream counter from 1 to 10 and granted valid signed playback URLs (`HTTP 200 OK`).

2. Viewer 11 requested an 11th distinct title. The API detected that the node had reached its cap ($10 \geq 10$). It immediately returned an `HTTP 429` error without contacting the home node:

```
HTTP/1.1 429 Too Many Requests
Content-Type: application/json
Retry-After: 30

{
  "error": "node_busy",
  "message": "Edge node has reached its maximum concurrent remote
   stream capacity (10 streams). Please try again shortly.",
  "retry_after_seconds": 30,
  "node_id": "node-4060-desktop"
}
```

Listing 6.1: Admission Control HTTP 429 Response

3. The web player displayed a clear “Node Busy” message with a 30-second retry countdown timer, protecting the home computer from network overload.

6.5 Streaming Latency and Offline Handling

Table 6.3 compares playback startup time and latency across the different playback routes.

Table 6.3: Playback Latency Across Streaming Routes

Streaming Route	**Time to First Byte (TTFB)**	**Buffer Startup Delay**	**Cloud Egress Cost**
Direct LAN (Mini-Jellyfin)	**3.2 ms**	**240 ms**	**₹ 0.00**
Remote Cache Hit	14.8 ms	680 ms	Included in VPS
Remote Cache Miss (First Ingest)	86.4 ms	1,180 ms	Included in VPS
Remote Uncached Fallback	94.2 ms	1,320 ms	Included in VPS

6.5.1 Node Disconnect and Heartbeat Recovery

To test offline detection (FR-9), we simulated an unexpected node shutdown (`kill -9`):

- At $t = 0$ s, the edge daemon stopped sending heartbeats.
- At $t = 15$ s, the expected heartbeat did not arrive.
- At $t = 20$ s, the Redis key `heartbeat:node-4060-desktop` expired automatically.
- The catalog immediately flagged the node as *Offline*, and public media cards displayed an amber "Node Offline" badge, preventing users from attempting to stream from an unavailable node.

6.6 Automated Code Quality and Test Results

To ensure code stability, all modules were tested through automated quality checks before release:

- **Go Unit and Integration Tests:** All 48 test suites across `apps/api` and `apps/edge` passed with race detection enabled (`go test -v -race -cover ./...`), achieving 81.4% average statement coverage with zero data races.
- **Frontend Code Quality:** Verified via strict TypeScript compilation and Next.js production builds (`npm run lint`, `tsc -noEmit`, and `next build`), completing with zero errors or warnings.
- **End-to-End System Health (`make demo-check`):** Verified PostgreSQL connectivity, Redis TTL key eviction, FRP reverse tunnel transport, and Origin Shield S3 operations with all-green operational status.

CHAPTER 7

Team Contributions and Division of Work

CustardFin was collaboratively designed, implemented, and validated by two MCA students: **Anish Ghadi** (Seat No.: **25P0320009**, Edge Lead) and **Anosh Chodankar** (Seat No.: **25P0320010**, Cloud Lead).

Summary of Individual Deliverables:

- **Anish Ghadi (Edge Tier Lead):** Home Edge subsystem (`apps/edge`), recursive scanning, SHA-256 hashing, managed FFmpeg (NVENC GPU probe with CPU fallback), embedded SQLite WAL, outbound FRP reverse tunnel daemon, and Mini-Jellyfin offline web UI.
- **Anosh Chodankar (Cloud Tier Lead):** Cloud Control Plane (`apps/api`) and Web Tier (`apps/web`), hexagonal Go REST API, PostgreSQL catalog schemas, Redis heartbeat tracking, Origin Shield S3 cache with HMAC-SHA256 signing, and Next.js 15 web player.

Subsystem Workload Distribution: Both authors jointly formulated the 10 System Invariants, designed the admission limiter, executed the 10-viewer load simulation, and co-authored documentation (Table 7.1).

Table 7.1: Subsystem Responsibility and Workload Contribution Matrix

Subsystem	Primary Deliverables	Primary Owner	Anish	Anosh
Edge Ingestion	Scanner, SHA-256 Hasher, SQLite WAL	Anish Ghadi	**85%**	15%
Transcoding Engine	Managed FFmpeg, NVENC GPU, CPU Fallback	Anish Ghadi	**85%**	15%
Reverse Tunnel	Outbound FRP Client Daemon, Auto-Reconnect	Anish Ghadi	**80%**	20%
Mini-Jellyfin Hub	Embedded Offline HTML5/JS Web Player	Anish Ghadi	**80%**	20%
Cloud Control API	Go REST API, Hexagonal Architecture, Slog	Anosh Chodankar	20%	**80%**
Database & Cache	PostgreSQL Schemas, Pgxpool, Redis TTL	Anosh Chodankar	15%	**85%**
Origin Shield Cache	S3 Pull-Through Cache, 24h LRU, HMAC Signer	Anosh Chodankar	20%	**80%**
Web Presentation	Next.js 15 App Router, HLS Player, HUD	Anosh Chodankar	15%	**85%**
Architecture & Tests	10 Invariants, Admission Router, Load Tests	Joint Collaboration	**50%**	**50%**
Overall Workload	**Full Distributed System Delivery**	**Equal Effort**	**50%**	**50%**

CHAPTER 8

Conclusion and Future Scope

Traditional video streaming platforms store massive video libraries in centralized cloud data centers, creating high recurring costs for storage and bandwidth. **CustardFin** solves this problem by keeping video files on the owner's home computer and using a lightweight cloud server only for catalog search, connection coordination, and temporary caching. This approach provides reliable video streaming, protects personal privacy, and reduces cloud operating costs by 95.89%.

8.1 Summary of Achievements

CustardFin demonstrates that personal media sovereignty and public community sharing can be unified without prohibitive cloud infrastructure expenses. The project successfully achieved all core objectives:

- **Zero Cloud Video Storage:** Master video binaries remain permanently on home storage. The cloud maintains only 650-byte JSON manifests, eliminating recurring cloud storage fees (Invariant 1, 8).
- **Residential CGNAT Traversal:** Outbound-only TLS reverse tunnels (FRP) allow home nodes behind residential broadband firewalls to serve streams globally without open router ports or static IPs (Invariant 2).
- **Hardware-Accelerated Transcoding:** Local NVENC pre-encoding achieves 10.74× real-time throughput (322.3 fps) with host CPU utilization below 5%, supported by seamless CPU software fallback (Invariant 6).
- **Origin Shield Bandwidth Offloading:** The 24-hour cloud cache offloads 90% of traffic during multi-viewer streaming, allowing a modest 20 Mbps residential uplink to support 10 concurrent viewers smoothly (Invariant 7, 10).
- **Empirically Verified Cost Reduction:** Monthly operating costs drop by **95.89%** from ₹ 1,00,380.00 in centralized cloud VOD to ₹ 4,128.60 with CustardFin, saving ₹ 11.55 Lakhs annually for a 100-user community.

8.2 Current Limitations (Version 1)

While CustardFin provides a fully functional streaming platform, several limitations exist in Version 1:

1. **Single Relay Server Dependency:** Uncached chunks must travel through a single cloud relay server running `frps`. If this relay goes down, remote playback for uncached chunks is temporarily interrupted.
2. **No Direct Browser P2P Swarming:** In Version 1, edge nodes stream through the cloud relay; direct peer-to-peer chunk sharing between browser viewers using WebRTC is not yet implemented.
3. **No Proprietary DRM:** To remain completely free and open-source, CustardFin relies on HMAC-SHA256 signed URLs and HLS segmentation rather than commercial DRM systems like Google Widevine or Apple FairPlay.
4. **Web-First Player:** The frontend is currently a responsive web application for desktop and mobile browsers. Dedicated native applications for smart TVs (Android TV, Apple TV) are not yet available.

8.3 Future Enhancements

The modular design of CustardFin allows for several planned enhancements:

- **Distributed Multi-Region Relays:** Deploying reverse tunnel relays across multiple cloud regions using GeoDNS will reduce latency for global viewers and eliminate single points of failure.
- **WebRTC Peer-Assisted Streaming:** Enabling viewers watching the same popular video to share cached chunks directly through browser WebRTC data channels will further reduce cloud bandwidth.
- **Cryptographic Manifest Verification:** Adding Ed25519 digital signatures to public manifests will verify node identity and metadata integrity.
- **Smart TV Applications:** Native apps built with Flutter or React Native can bring CustardFin to smart TVs with automatic local network discovery.

8.4 Concluding Remarks

CustardFin proves that high-quality media streaming does not require expensive, centralized cloud storage. By keeping video files safely on the creator's own computer and using a lightweight cloud server for discovery and temporary caching, CustardFin offers a private, community-focused, and highly economical streaming platform. This project successfully fulfills all academic and practical requirements of the MCA Cloud Computing Laboratory curriculum.

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